

A strongly forward isometric noncontractive representation of the finite Leavitt algebra

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Status. Claimed full negative resolution (counterexample), likely valid pending human review. The construction works for every finite $d \geq 2$ and every $1 \leq p < \infty$.

1 Source question

For $d \geq 2$, the Leavitt algebra L_d is the universal unital complex algebra with generators $s_1, \dots, s_d, t_1, \dots, t_d$ satisfying

$$t_j s_k = \delta_{jk} 1, \quad \sum_{j=1}^d s_j t_j = 1. \quad (1)$$

Phillips calls a representation $\rho : L_d \rightarrow L(E)$ *contractive on generators* if all $\rho(s_j)$ and $\rho(t_j)$ have norm at most one. It is *strongly forward isometric* if every $\rho(s_j)$ is an isometry and, for every $\lambda \in \mathbb{C}^d$, the operator $\rho(s_\lambda) = \sum_j \lambda_j \rho(s_j)$ is a scalar multiple of an isometry [1, Definition 2.12].

After giving a strongly forward isometric representation of L_∞ which is not contractive on generators, Phillips asks whether the same behavior can occur for finite L_d [1, Remark 2.13 and p. 15].

2 Idea of the proof

The standard representation of L_d on ℓ^p identifies ℓ^p with the p -direct sum of d copies of itself: its forward generators insert a vector into disjoint coordinate classes, and its reverse generators extract those classes. We apply a nonisometric change of basis to the first two copies.

Choose the change of basis so that each new column has ℓ^p -norm one. This makes every new forward generator an isometry. More strongly, any linear combination of the forward generators still has a norm-scaling factor independent of the input vector, so it is a scalar multiple of an isometry. But the inverse change of basis has a row of norm greater than one. The corresponding reverse generator is therefore expansive. The algebraic Leavitt relations are preserved because the changes on the forward and reverse sides are inverse matrices.

3 The counterexample

Theorem 1. *For every $d \geq 2$ and every $1 \leq p < \infty$, there exists a strongly forward isometric representation*

$$\rho : L_d \longrightarrow L(\ell^p(\mathbb{N}))$$

In Definition 7.4 and Definition 7.6 we will see further conditions on representations which are natural when $E = L^p(X, \mu)$.

Definition 2.12. Let A be any of L_d (Definition 1.1), C_d (Definition 1.2) or L_∞ (Definition 1.3). Let E be a nonzero Banach space, and let $\rho: A \rightarrow L(E)$ be a representation.

- (1) We say that ρ is *contractive on generators* if for every j , we have $\|\rho(s_j)\| \leq 1$ and $\|\rho(t_j)\| \leq 1$.
- (2) We say that ρ is *forward isometric* if $\rho(s_j)$ is an isometry for every j .
- (3) We say that ρ is *strongly forward isometric* if ρ is forward isometric and (following Definition 1.13) for every $\lambda \in \mathbb{C}^d$, the element $\rho(s_\lambda)$ is a scalar multiple of an isometry.

Remark 2.13. A representation which is contractive on generators is clearly forward isometric.

A representation of L_d which is contractive on generators need not be strongly forward isometric. See Example 3.11 below. We will see in Example 3.5 below that a strongly forward isometric representation of L_∞ need not be contractive on generators. We do not know whether this can happen for L_d with d finite.

We now describe several ways to make new representations from old ones. The

Figure 1: The definitions and open question in [1, p. 10].

which is not contractive on generators. Consequently, the finite- d question in [1, Remark 2.13] has a negative answer.

Proof. Let $V_j, W_j \in L(\ell^p(\mathbb{N}))$, $1 \leq j \leq d$, be the standard spatial Leavitt family. Explicitly,

$$V_j \delta_n = \delta_{d(n-1)+j}, \quad (W_j \xi)(n) = \xi(d(n-1) + j).$$

Thus

$$W_j V_k = \delta_{jk} I, \quad \sum_{j=1}^d V_j W_j = I, \tag{2}$$

and the ranges of $V_1, \dots, V_d$ have disjoint supports. In particular, for all $a = (a_1, \dots, a_d) \in \mathbb{C}^d$ and $\xi \in \ell^p$,

$$\left\| \sum_{j=1}^d a_j V_j \xi \right\|_p = \left(\sum_{j=1}^d |a_j|^p \right)^{1/p} \|\xi\|_p. \tag{3}$$

Set $\alpha = 2^{-1/p}$ and define

$$\begin{aligned} S_1 &= V_1, & S_2 &= \alpha(V_1 + V_2), & S_j &= V_j \quad (3 \leq j \leq d), \\ T_1 &= W_1 - W_2, & T_2 &= \alpha^{-1} W_2, & T_j &= W_j \quad (3 \leq j \leq d). \end{aligned}$$

The relations $T_j S_k = \delta_{jk} I$ involving indices at least three follow immediately from (2). For the first two indices,

$$\begin{aligned} T_1 S_1 &= (W_1 - W_2) V_1 = I, & T_1 S_2 &= \alpha(W_1 - W_2)(V_1 + V_2) = 0, \\ T_2 S_1 &= \alpha^{-1} W_2 V_1 = 0, & T_2 S_2 &= W_2(V_1 + V_2) = I. \end{aligned}$$

Moreover,

$$\begin{aligned} S_1 T_1 + S_2 T_2 &= V_1(W_1 - W_2) + (V_1 + V_2)W_2 \\ &= V_1 W_1 + V_2 W_2. \end{aligned}$$

Adding $\sum_{j=3}^d V_j W_j$ gives the identity. Hence the S_j, T_j satisfy (1), and the universal property produces a representation ρ with $\rho(s_j) = S_j$ and $\rho(t_j) = T_j$.

We next verify strong forward isometry. Formula (3) shows that S_1 is an isometry and, since $2\alpha^p = 1$, that S_2 is an isometry; the remaining S_j are already isometries. For $\lambda = (\lambda_1, \dots, \lambda_d) \in \mathbb{C}^d$,

$$\sum_{j=1}^d \lambda_j S_j = (\lambda_1 + \alpha \lambda_2) V_1 + \alpha \lambda_2 V_2 + \sum_{j=3}^d \lambda_j V_j.$$

Therefore (3) gives, for every $\xi \in \ell^p$,

$$\left\| \sum_{j=1}^d \lambda_j S_j \xi \right\|_p = c(\lambda) \|\xi\|_p, \quad (4)$$

where

$$c(\lambda) = \left(|\lambda_1 + \alpha \lambda_2|^p + |\alpha \lambda_2|^p + \sum_{j=3}^d |\lambda_j|^p \right)^{1/p}.$$

The triangular change of variables appearing here is invertible, so $c(\lambda) > 0$ whenever $\lambda \neq 0$. Dividing the operator in (4) by $c(\lambda)$ gives an isometry. (For $\lambda = 0$ the operator is the zero scalar multiple of any isometry.) Thus ρ is strongly forward isometric.

Finally, W_2 has norm one, and hence

$$\|\rho(t_2)\| = \|T_2\| = \alpha^{-1} = 2^{1/p} > 1.$$

Thus ρ is not contractive on generators. □

4 Matrix interpretation and verification

On the first two coordinates the construction is the generator change

$$B = \begin{pmatrix} 1 & \alpha \\ 0 & \alpha \end{pmatrix}, \quad B^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & \alpha^{-1} \end{pmatrix}.$$

The columns of B have ℓ_2^p -norm one, which is exactly forward isometry. The second row of B^{-1} has norm $\alpha^{-1} > 1$, which is exactly the failed reverse-generator contraction. This is also a quick audit of why both properties can coexist without contradicting the Leavitt relations.

No numerical or computer-assisted step is used. The proof consists only of the displayed operator identities and the disjoint-support norm formula (3).

5 Scope and novelty check

The construction covers all finite $d \geq 2$ and all $1 \leq p < \infty$, so it fully answers the existence question as stated. It makes no assertion about spatiality, closures of representation ranges, or isomorphism classes of the resulting completed Banach algebras.

A bounded search on 9 August 2026 covered the run registry, solution, attempt, and proof-gap indexes; exact searches for “strongly forward isometric” with “contractive on generators”; the verbatim open sentence; title/citation searches for arXiv:1201.4196; and searches for finite- d Leavitt representations. The search found the source and later general L^p -operator-algebra literature, but no later answer to this exact question. This supports, but does not certify, novelty.

Recommended human review is high priority but narrow: check the four first-two-coordinate relations, the reconstruction of the identity, and the norm formula (4).

References

- [1] N. Christopher Phillips, *Analogues of Cuntz algebras on L^p spaces*, arXiv:1201.4196v1, 2012.